

The following shows an example of how to solve a system of equations with three variables in modular arithmetic. This is useful in cracking certain ciphers or pseudorandom number generators, which are important to cryptography. Many of the steps are similar to solving a system of equations in standard arithmetic, but there are differences, especially in the usage of division.

Example

We are given the equations:

$$\begin{aligned}3a + 4b + c &= 9(\text{mod } 17) \\4a + 9b + c &= 13(\text{mod } 17) \\9a + 13b + c &= 2(\text{mod } 17)\end{aligned}$$

and we want to solve for a, b, c . As we would in standard arithmetic. We will first eliminate c by subtracting the first equation from the second, and the second from the third. So

$$\begin{aligned}(4 - 3)a + (9 - 4)b + (c - c) &= (13 - 9)(\text{mod } 17) \\(9 - 4)a + (13 - 9)b + (c - c) &= (2 - 13)(\text{mod } 17)\end{aligned}$$

which simplify to

$$\begin{aligned}a + 5b &= 4(\text{mod } 17) \\5a + 4b &= -11(\text{mod } 17)\end{aligned}$$

Next, let's eliminate b . To do so, first we multiply the top equation by 4, and the bottom equation by 5.

$$\begin{aligned}4a + 20b &= 16(\text{mod } 17) \\25a + 20b &= -55(\text{mod } 17)\end{aligned}$$

and then we can subtract the second equation from the first, to get

$$(4 - 25)a + (20 - 20)b = 16 - (-55)(\text{mod } 17)$$

which simplifies to

$$(-21)a = 71(\text{mod } 17)$$

or

$$13a = 3(\text{mod } 17)$$

This is because $-21 = 13 \pmod{17}$ and $71 = 3 \pmod{17}$. Now, we want to divide both sides by 13. In modular arithmetic, that means multiplying both sides by the *modular inverse* of 13 $\pmod{17}$. We find that the inverse of 13 mod 17 is 4. Thus,

$$\begin{aligned} 13 * 4a &= 3 * 4 \pmod{17} \\ a &= 12 \pmod{17} \end{aligned}$$

Now, as with system of equations in standard arithmetic, we substitute the value we have found for a back into one of the previous equations to solve for b . So, using the equation $a + 5b = 4 \pmod{17}$ from before,

$$\begin{aligned} (12) + 5b &= 4 \pmod{17} \\ 5b &= 4 - 12 \pmod{17} \\ 5b &= 9 \pmod{17} \end{aligned}$$

And since the modular inverse of 5 $\pmod{17}$ is 7,

$$\begin{aligned} 5 * 7b &= 7 * 9 \pmod{17} \\ b &= 12 \pmod{17} \end{aligned}$$

And, finally, using the very first equation, $3a + 4b + c = 9 \pmod{17}$

$$\begin{aligned} 3(12) + 4(12) + c &= 9 \pmod{17} \\ c &= 9 - 84 \pmod{17} \\ c &= 10 \pmod{17} \end{aligned}$$

and so, we have found that a, b, c are 12, 12, 10 respectively.